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Department of Information Technology

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Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

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Experiment – 8: Security as a Service (SECaaS)

Aim: To study and Implement Security as a Service on AWS using ADDS service.
(Free Tier Cloud Platforms).

Objective: After performing the experiment, the students will be able to –

- to understand Cloud security at AWS
- AWS Directory Service for identity management
- deploy AWS administration and security services.

Lab objective mapped: ITDO8024.4: To learn cloud deployment model and experiment security services over different cloud platforms.

Prerequisite:

An AWS Account

Requirements: Cloud Login, Desktop, Browser, Internet etc.

Pre-Experiment Theory:

AWS Directory Service

AWS Directory Service is a managed service that provides directories that contain information about your organization, including users, groups, computers, and other resources.

Windows Server Manager

Microsoft Windows Server Manager is a tool to view and manage server roles and make configuration changes. Server Manager allows administrators to manage local and remote servers without requiring physical access to the servers or enabling Remote Desktop Protocol connections.

Procedure

AWS Managed Microsoft AD prerequisites

- To create a AWS Managed Microsoft AD directory, you need a VPC with the following:
- At least two subnets. Each of the subnets must be in a different Availability Zone.
- The VPC must have default hardware tenancy.
- You cannot create a AWS Managed Microsoft AD in a VPC using addresses in the 198.18.0.0/15 address space.

To create an AWS Managed Microsoft AD directory

1. AWS Directory Service console navigation pane, choose Set up directory.
2. Select directory type : choose AWS Managed Microsoft AD, and then click Next.
3. Enter directory information page :
4. Edition :Choose Standard Edition.
5. Directory DNS name :The fully qualified name (FQDN) for the directory, *beit_exp5.com*.
6. Directory NetBIOS name : optional.
7. Directory description : An optional description for the directory.
8. Admin password : The password for the directory administrator. The directory creation process creates an administrator account with the user name **Admin**. (The password cannot include the word "admin."). Read rules for setting up the directory administrator password.
9. Confirm password
10. Choose VPC and subnets page ([refer to VPC creation procedure](#)) :
11. The VPC for the directory.
12. Subnets: Choose the subnets for the domain controllers. The two subnets must be in different Availability Zones.
13. choose Next.
14. On the Review & create page : review the directory information. Choose Create directory.
15. *Creating the directory takes 20 to 40 minutes (What [gets](#) created).*
16. *Once created, the Status value changes to Active (green in color).*

To seamlessly join a Windows EC2 instance

1. Go to Amazon EC2 console
2. From the Region selector in the navigation bar, choose the same Region as the existing directory.
3. Launch Instance.
4. AMI : win server 2019 free tier full edition
5. instance type : t2.micro
6. Next: Configure Instance Details.
7. Step 3 page:
8. Network : VPC that your directory was created in.
9. Subnet : choose one of the public subnets in your VPC. (The subnet that you choose must have all external traffic routed to an internet gateway.)
10. Domain join directory : choose your domain from the list.
11. IAM role:
 - a. Choose Create role.

- b. Under Select type of trusted entity, choose AWS service.
 - c. service role will use : choose EC2.
 - d. Select your use case : choose EC2
 - e. Next: Permissions.
 - f. In the list of policies : 1. AmazonSSMManagedInstanceCore
2. AmazonSSMDirectoryServiceAccess policies
 - g. Choose Next: Tags(Optional)
 - h. choose Next: Review.
 - i. Role name : EC2DomainJoin
 - j. Choose Create role.
12. Go back to the Step 3 page. For IAM role, choose the refresh icon next to IAM role. Choose created role and leave the rest of the settings with default values
 13. choose Next: Add Storage.
 14. On both the Step 4 and Step 5 pages, leave the default settings or make changes as needed, and then choose the Next buttons.
 15. On the Step 6 page, select a security group for the instance that has been configured to allow remote access to the instance from your network, and then choose Review and Launch.
 16. Step 7 page, choose Launch, select a key pair, and then choose Launch Instance..

To install the Active Directory administration tools on Windows Server 2012 through Windows Server 2019

1. Open Server Manager from the Start screen by choosing Server Manager.
2. In the Server Manager Dashboard, choose Add roles and features. In the Add Roles and Features Wizard choose Installation Type, select Role-based or feature-based installation, and choose Next.
3. Under Server Selection, make sure the local server is selected, and choose Features in the left navigation pane.
4. In the Features tree, open Remote Server Administration Tools, Role Administration Tools, select AD DS and AD LDS Tools, scroll down and select DNS Server Tools, and then choose Next.
5. Review the information and choose Install.
6. When the feature installation is finished, the Active Directory Domain Services and Active Directory Lightweight Directory Services Tools are available on the Start screen in the Administrative Tools folder.

To create a user

1. Open the Active Directory Users and Computers tool. There is a shortcut to this tool in the Administrative Tools folder.
2. On the Action menu, click New, and then click User to open the new user wizard.
3. On the first page of the wizard, enter the values for the following fields, and then click Next.

4. First name : student name
5. Last name : student last name
6. User logon name : first name
7. type a temporary password in Password and Confirm Password. Make sure the User must change password at next logon option is selected. None of the other options should be selected. Click Next.
8. On the third page of the wizard, verify that the new user information is correct and click Finish.
9. The new user will appear in the Users folder.

Create a new VPC

1. VPC console
2. choose Launch VPC Wizard.
3. Under Step 1: Select a VPC Configuration, on VPC with a Single Public Subnet, choose Select.
4. Enter the following information into the wizard and choose Create VPC.

IP CIDR block

10.100.0.0/16

VPC name

BEIT_EXP5_VPC

Public subnet

10.100.0.0/24

Availability Zone

us-east-1a

Subnet name

exp5_Subnet1

Enable DNS hostnames

Leave default selection

Hardware tenancy

Default

It takes several minutes for the VPC to be created.

Post-Experiments Exercise

1. Name three directory types that AWS supports? (to be written in hand)
2. List any five permissions an Admin account has to perform the administrative activities for your organizational unit (OU)? (to be written in hand)

Extended Theory:

What is Windows Server Manager?(soft copy)

Ans: Windows Server Manager is a built-in administrative tool in Microsoft Windows Server designed to help system administrators manage, monitor, and configure servers efficiently from a single, centralized interface. It was introduced to simplify server management tasks and reduce the need to use multiple separate tools for different configurations. At its core, Server Manager provides a **dashboard view** that displays important information about the server's health, performance, and current status. Administrators can quickly see alerts, system events, and notifications, making it easier to detect and resolve issues without manually checking each component. In summary, Windows Server Manager is a powerful and user-friendly tool that centralizes server administration, improves efficiency, supports remote management, and helps maintain the health and performance of multiple servers in an organized way.

Results/Calculations/Observations:

Post Experimental Exercise-

How to reset password for already created user in AWS AD service?

Conclusion:

1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:

- [1] [Online]<https://learn.microsoft.com/en-us/azure/app-service>
[2] [Online]<https://learn.microsoft.com/en-us/azure/app-service/quickstart-nodejs>

The screenshot displays the AWS Management Console interface for VPCs. At the top, there's a header 'Your VPCs' with tabs for 'VPCs' and 'VPC encryption controls'. Below this, a sub-header 'Your VPCs (1/5)' is followed by a search bar and a table of VPCs. The table has columns for Name, VPC ID, State, Encryption, and Block Public. The 'riya_vpc' is selected, and its details are shown in a panel below the table. The details panel has tabs for 'Details', 'Resource map', 'CIDRs', 'Flow logs', 'Tags', and 'Integrations'.

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Publi
my-vpc-1	vpc-0e7cd334d4aa33a04	Available	-	-	Off
vpc1alister	vpc-0024641677569d426	Available	-	-	Off
riya_vpc	vpc-036ec16721e305a71	Available	-	-	Off
-	vpc-02221dd13ad9df729	Available	-	-	Off

vpc-036ec16721e305a71 / riya_vpc

Details | Resource map | CIDRs | Flow logs | Tags | Integrations

gateways

igw-0a04514a45602a255

Internet gateway igw-0a04514a45602a255 successfully attached to vpc-036ec16721e305a71

Notifications 0 0 2 0 0 0

igw-0a04514a45602a255 / riya_gateway

Details

Internet gateway ID
igw-0a04514a45602a255

State
Attached

VPC ID
vpc-036ec16721e305a71 | riya_vpc

Owner
780899468016

Tags (1)

Search tags

Manage tags

Key	Value
Name	riya_gateway

